

Checklist

What you should know

After studying this subtopic you should be able to:

- Deduce the Brønsted—Lowry acid and base in a reaction.
- Deduce the formula of the conjugate acid or base of any Brønsted—Lowry base or acid.
- Interpret and formulate equations to show acid—base reactions of amphoteric species.
- Perform calculations involving the logarithmic relationship between pH and $[H^+]$.
- Recognise solutions as acidic, neutral and basic from the relative values of $[H^+]$ and $[OH^-]$.
- Recognise that acid—base equilibria lie in the direction of the weaker conjugate.
- Formulate equations for the reactions between acids and metal oxides, metal hydroxides, hydrogencarbonates and carbonates.
- Sketch and interpret the general shape of the pH curve of a strong acid and strong base neutralisation reaction.

Higher level (HL)

- Interconvert $[H^+]$, $[OH^-]$, pH and pOH values.
- Write K_a and K_b expressions.
- Interpret the relative strengths of acids and bases from the K_a , K_b , pK_a or pK_b values.
- Solve problems involving K_a and K_b values.
- Construct equations for the hydrolysis of ions in a salt, and predict the effect of each ion on the pH of the salt solution.
- Interpret the general shapes of pH curves for all four combinations of strong and weak acids and bases.
- Construct equilibria expressions to show why the colour of an indicator changes with pH.
- Identify an appropriate indicator for a titration from the identity of the salt and the pH range of the indicator.

- Describe the composition of acidic and basic buffers and explain their actions.
- Solve problems involving the composition and pH of a buffer solution, using the equilibrium constant.

Practical skills

Once you have completed this subtopic, go to Practical 1.1.8a: Acid—base titration with a chemical indicator (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-titration-with-a-chemical-indicator-8a-id-46724/>) and Practical 1.1.8b: Acid—base titration with a pH sensor (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-titration-with-a-ph-sensor-8b-id-46725/>) in which you can practise your titration skills.